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10. The Ecosystem: Extending React Native

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Civilization advances by extending the number of operations which we can perform without thinking about them.

—Alfred North Whitehead

If you have come this far, you can proudly say you have become a React Native developer. In this final chapter you learn how expedite your React Native development by using some very useful, stable, and popular libraries. These are designed to make your life a bit easier and help you create your apps faster. This chapter covers the following topics:

- Popular React Native libraries

- Community, Help, and where to go from here

Popular React Native Libraries

From the time of its inception the React Native ecosystem has grown by leaps and bounds. The React Native community is vibrant and exceptionally productive: With every passing week, something new is always coming up to untangle the complications of development. By the time you have reached this chapter and we have completed this book, a lot more must have happened (later in this chapter we share ways to stay updated with the community). However, this chapter provides a curated list of libraries organized based on categories to help you increase the velocity of your React Native development.

UI

Styled-components

Styled-components allows you to write actual CSS code to style your components. It removes the mapping between components and styles: Using components as a low-level styling construct makes it easy. See <https://github.com/styled-components/styled-components>.

Lottie-react-native

Lottie is a mobile library for Android and iOS that parses Adobe After Effects animations exported as JSON with body-movin (an After Effects extension to export animations for the Web) and renders them natively on mobile platforms. Access the Lottie mobile library here: <https://github.com/react-native-community/lottie-react-native>.

React-native-vector-icons

This library is perfect for buttons, logos, and navigation and tab bars. It is easy to extend, style, and integrate into your project. It provides customizable icons for React Native with support for NavBar/TabBar/Toolbar, image source, and full styling. See <https://github.com/oblador/react-native-vector-icons>.

Forms

Formik

Formik is a simple library that helps you with the three parts that make forms in React complicated: getting values in and out of form state, validation and error messages, and handling form submission. See <https://github.com/jaredpalmer/formik>.

Redux-form

`Redux-form` is the most convenient way to manage a form state in Redux. To make use of this module you must have some idea about the Redux state container and higher order components.

See <https://github.com/erikras/redux-form/>.

Type Checking and Linting

ESLint

ESLint is an open source project that has as its ultimate goal to provide a pluggable linting utility for JavaScript. There are many popular ESLint configurations available from popular projects that can import for your application while also creating new custom linting rules based on your requirement. See

<https://eslint.org/>.

Prop-types

Prop-types is a library that helps in runtime type checking for React props and similar objects. See

<https://www.npmjs.com/package/prop-types>.

Flow

Flow is a static type checker for JavaScript that helps identify problems with your code early instead of guessing and checking. Flow provides real-time feedback as you code and make your changes. See <https://flow.org/>.

Testing

Jest

Jest is a testing framework that is simple to use and integrate with your React Native application. It comes out of the box with React Native versions 0.38 and above. Jest also allows for snapshot testing, which is a brilliant way to manage changes in the UI. See `url`: <https://jestjs.io/>.

Enzyme

Enzyme is a testing tool that was created and open sourced by Airbnb. It supports tons of features like shallow rendering, full DOM rendering, and static rendered markup. It is a great add-on, along with Jest. Enzyme APIs are intuitive and flexible as they imitate JQuery APIs for DOM manipulations. See <https://github.com/airbnb/enzyme>.

Chai

Chai is an assertion testing library based on test-driven and behavior-driven development. Just like Enzyme, Chai also enhances other testing frameworks. See

<https://www.chaijs.com/>.

Mocha

Mocha is a JavaScript testing framework that helps make asynchronous testing simple. Mocha runs test serially and provides accurate reporting, while mapping uncaught exceptions to the correct test cases. See <https://mochajs.org/>.

Interacting with APIs and Back End

Axios

Axios is an HTTP client for JavaScript that helps make HTTP requests to REST endpoints and perform CRUD operations. Axios supports Promise API, intercept request and response, helps transform request and response data, and has many more features. See <https://github.com/axios/axios>.

Apollo

If you plan to use GraphQL you will end up using Apollo, which is an implementation of GraphQL that helps manage data in the cloud. Apollo includes two open source libraries for

the client and server, in addition to developer tools that provide everything you need to run a graph API in production with confidence. See

<https://www.apollographql.com/docs/react/recipes/react-native.html>.

React-native-firebase

React-native-firebase is a collection of official React Native modules connecting you to Firebase services; each module is a lightweight JavaScript layer connecting you to the native Firebase SDKs for both iOS and Android. See

<https://github.com/invertase/react-native-firebase>.

Routing

React Router

React Router is a collection of navigational components that compose declaratively with your application. Whether you want to have URLs that can be bookmarked for your web app or a composable way to navigate in React Native, React Router works perfectly. See <https://reacttraining.com/react-router>.

React Navigation

React Navigation was born from the React Native community's need for an extensible yet easy-to-use navigation solution written entirely in JavaScript (so you can read and understand all of the source), on top of powerful native primitives. See

<https://reactnavigation.org/>.

Utilities

Lodash

Lodash is a JavaScript library that provides utility functions for common programming tasks using the functional programming paradigm. Lodash is the most commonly used library in any application and it is very popular in the JavaScript world.

See <https://lodash.com/docs/4.17.11>.

Ramda

Ramda is a library designed specifically for a functional programming style, one that makes it easy to create functional pipelines and never mutates user data. See

<https://ramdajs.com/>.

Moment

`Moment.js` is brilliant for managing dates in JavaScript, which is something you will always stumble on when developing an

application. See <https://momentjs.com/>.

Reselect

Reselect is a simple “selector” library with Redux. Having key features like selectors can compute derived data, allowing Redux to store the minimal possible state. Selectors are efficient; a selector is not recomputed unless one of its arguments changes. They are also composable, and they can be used as input to other selectors. See

<https://github.com/reduxjs/reselect>.

Validate.js

Validation is part of any application. `Validate.js` serves this purpose by providing a declarative way of validating JavaScript objects. With `Validate.js`, validation constraints can be declared in JSON and shared between clients and the server. See <https://validatejs.org/>.

React-native-device-info

This is a simple library, and as its name suggests, it provides device information for React Native for iOS and Android. It has a long list of APIs to provide in-depth information about the device on which an application is running. See

<https://github.com/rebeccahughes/react-native-de-vice-info>.

Where to Get Help

This section provides some suggestions on where to get help in the React Native community.

React Native Repository

The React Native repository is maintained by a full-time Facebook React Native core team, but there is huge community that is always contributing to keeping this framework stable. You can always raise an issue if you find one with the framework in the GitHub repository, and there you can also find solutions to past issues. See

<https://github.com/facebook/react-native/issues>. To report a bug in the framework you can use the bug report format available at https://github.com/facebook/react-native/issues/new?template=bug_report.md.

Stack Overflow

Stack Overflow is a place where people across the React Native community help each other. You can post questions and get answers pretty quickly. You can also help fellow developers as

you proceed in your journey toward becoming an expert in React Native (or any other technology). By giving answers to various questions, your score on Stack Overflow increases, which is a kind of motivation for helping others. Many developers actually boast about their Stack Overflow stats. See <https://stackoverflow.com/questions/tagged/react-native?sort=frequent> for a list of existing questions on React Native. To ask a question with a React Native tag, go to <https://stackoverflow.com/questions/ask?tags=react-native>.

Stay Updated with React Native

You should also stay in tune with the latest happenings in React Native with the official documentation available at <https://github.com/facebook/react-native-website>. The official blog of React Native maintained at <https://facebook.github.io/react-native/blog/> will keep you updated on what is new. You can also connect with the official React Native Twitter account, which keeps updated with both React Native and Reactjs. See <https://twitter.com/reactjs>.

React Native Communities

Sometimes if you don't get an answer quickly on GitHub issues or Stack Overflow, it's a good idea to get in touch with the larger community instantly. For that you can join the React Discord channel and connect with fellow developers. Incidentally, it's not necessary for you have questions; you can always share your discovery or maybe your next open source React Native project. This is a good way to get some visibility. See <https://discordapp.com/invite/0ZcbPKXt5bZjGY5n>.

Another way to interact with the React Native developer community is to be part of various online groups and forums. Here are a few recommended ones you can join:

- *React Native Spectrum:* <https://spectrum.chat/react-native>
- *React Native Facebook group:* <https://www.facebook.com/groups/react.native.community>
- *Expo forum:* <https://forums.expo.io/>

Knowledge

Besides the official React Native blog, there are some amazing posts written on other blogs by community members that can further enhance your knowledge. Here are two recommended ones:

- *DevTo community:* <https://dev.to/t/reactnative>

- *React Native on Medium:* <https://medium.com/tag/react-native>

Discussions and Proposals

React Native, although very powerful, is still a young framework and its core team is always looking for great proposals, improvements, and discussions. Be part of this problem-solving effort and contribute to this thriving community. You can do this by following the formal channel at

<https://github.com/react-native-community/discussions-and-proposals>.

Summary

Now we have reached the end of our book and our last summary. In this chapter we provided information about various React Native libraries that can expedite our development time and give access to the enormous treasure trove of features built over time by the React Native community. You also learned about how to stay updated on this fast-moving framework by getting information from the right sources.

Although you have learned a lot during the course of this book, to truly master this topic you have to keep practicing and creating apps. There is no better way to become an expert at a

technology than learning on your own in a real-world scenario. You can contribute to the developer community by creating a module that still does not exist or by contributing to existing open source React Native repos. We are very excited about React Native, just like you, and look forward to seeing your work making a mark in the mobile development and React Native world.